

STANLEY CHEGE THUITA

OPEN-SOURCE BITCOIN PROTOCOL ENGINEER & SYSTEMS DEVELOPER

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MISSION & TARGET FOCUS

OPEN SOURCE GRANTS • FELLOWSHIPS • PROTOCOL DEVELOPMENT

Systems developer and open-source Bitcoin engineer with formal mathematical training from the University of Nairobi. Specializes in zero-dependency Go architecture, Lightning Network Daemon (LND) gRPC integrations, and sovereign financial infrastructure. Graduate of the Btrust Builders / Africa Free Routing / HRF Lightning Bootcamp and co-founder of BitDevs Kisumu. Focused on full-time upstream Bitcoin Core and Lightning protocol development through open-source developer grants (Brink, Btrust, OpenSats, Spiral) and research fellowships.

TECHNICAL CAPABILITIES

ATS-OPTIMIZED CATEGORIZATION

- Bitcoin & Lightning Protocol:** Lightning Network Daemon (LND), LND gRPC client architecture (`SendPaymentSync` , `FeeLimit_Fixed`), BOLT-11 HRP invoice parser, LNURL-pay RFC resolution, Regtest cluster automation, `bitcoin-cli` , `OP_RETURN` transaction anchoring, Cashu / Fedimint architectures.
- Systems & Go Architecture:** Go (Zero-Dependency Standard Library Architecture, Goroutines, Channels, `sync.Mutex` / `RWMutex` , `context` , native `net/http` multiplexing, `embed.FS`), SQL (PostgreSQL), Bash/Shell scripting, Linux systems administration, Docker containerization.
- Applied Mathematics & Cryptography:** Elliptic curve cryptography (`secp256k1`), Schnorr signatures (BIP-340), Taproot key path & script tree commitment (BIP-341), RFC 6238 TOTP, AES-256-GCM AEAD, SHA-256, Ed25519 digital signatures, finite fields, graph pathfinding (Dijkstra).
- Protocol Leadership & Tooling:** BitDevs Socratic seminar curation, technical documentation, Git upstream PR workflow (rebase, squash, code review), TTCN-3 test automation toolchains.

OPEN SOURCE PROTOCOL CONTRIBUTIONS & UPSTREAM PULL REQUESTS

VERIFIABLE IMPACT

minimato/payment-channels — Open Source Protocol Contributor

Merged PRs #23, #24, #30 | June 2026 – Present

- Implemented** portable payment channel schemas mapping East African ([PR #23](#)) and Southern African ([PR #24](#)) cash corridors using UN M49 geographical classifications and ISO 3166-1/4217 currency standards.
- Authored** Ethiopian Telebirr payment channel specification ([PR #22](#)) with E.164 phone number validation and explicit automation enum separation (`PaymentChannelAutomation.Manual` vs enterprise payout models).
- Engineered** schema extensions for Western Sahara cash channels ([PR #30](#)) and continental cash corridor registries utilizing centralized factory constructors.

Btrust Builders & Open Source Repositories — Protocol Contributor

Merged PRs #139, #186, #133 | June 2026 – Present

- Completed** intensive Addis Ababa Lightning Bootcamp sponsored by Btrust Builders, Africa Free Routing, and Human Rights Foundation (HRF), training on Lightning Network internals, routing gossip, and channel management.
- Contributed** to African developer milestone registries and open-source portfolio repositories across [PR #139](#), [PR #186](#), and [PR #133](#).
- Authored** merged sensor ingestion and telemetry engine patches for Lake Victoria water monitoring platform (`chemichemi`, [PR #1](#), [PR #5](#)) and Windows CI test automation for TTCN-3 toolchains (`nokia/ntt` [PR #778](#)).

BITCOIN PROOF OF WORK & PRODUCTION CODEBASES

DEPLOYED SYSTEMS

BitDevs Kisumu Chapter — Co-Founder & Technical Facilitator

April 2026 – Present | bitdevs-kisumu.vercel.app

- Co-founded** Western Kenya's Bitcoin developer community; organized Socratic Seminar #1 at LakeHub Kisumu, facilitating technical discussions on Bitcoin Core PRs, BIPs, and Layer-2 protocol mechanics under the Chatham House Rule.
- Engineered** zero-JavaScript community documentation portal with Astro and Tailwind CSS; published research on Blockstream Satellite 2.0 dish alignment (Azimuth 82.4°, Elevation 54.8°) for offline Lake Victoria node synchronization.

YeboBank & Yebente — Founder & Lead Systems Architect

June 2026 – Present | yebobank.vercel.app

- Architected** pure-Go community banking engine (`YeboBank`) and settlement layer (`Yebente`) executing dual-rail fiat-to-Lightning payment bridging across East African mobile money corridors.
- Implemented** custom LND gRPC client with `MacaroonCredential` PerRPCCredentials authentication, TLS x509 cert pool validation, and bounded fee limits (`FeeLimit_Fixed`) executing synchronized payments via `SendPaymentSync` .
- Engineered** custom BOLT-11 HRP invoice decoder parsing multiplier prefixes (`m` , `u` , `n` , `p`) into satoshis and HTTP client resolving LNURL-pay RFC endpoints with deterministic SHA-256 invoice hashing.

ProofChain & Authentication — Applied Cryptography & Bitcoin Core Regtest Automation

August 2026 – Present | github.com/altradits

- Automated** local Bitcoin Core Regtest testbeds using shell automation and `bitcoin-cli` , scripting 101-block coinbase maturation and anchoring SHA-256 certificate hashes into blockchain transactions via `OP_RETURN` script outputs.
- Implemented** RFC 6238 Time-Based One-Time Password (TOTP) from first principles in Go using `crypto/hmac` , `crypto/sha1` , big-endian binary buffers, dynamic 4-bit truncation modulo 10^6 , and AES-256-GCM authenticated symmetric encryption.

TouchType & AltSocial — Zero-Dependency Go Systems

April 2026 – Present | touchtype-72i6.onrender.com

- Engineered** high-concurrency web services in pure Go standard library with zero external dependencies, leveraging `sync.Mutex` state synchronization, native `net/http` multiplexers, and `embed.FS` asset compilation.
- Maintained** automated cross-compilation toolchains producing standalone single-binary distributions across Darwin, Linux, and Windows architectures (amd64, arm64).

ENGINEERING APPRENTICESHIP

EDUCATION & MATHEMATICS

Zone01, Kisumu

April 2026 – Present

Software Engineering Apprentice

- Completed 158 structured Go challenges mastering concurrency, pointers, channels, and network protocols.
- Maintained strict Git hygiene, branch rebasing, and peer review rigor across distributed microservice and CLI projects.
- Configured Docker containerization across Linux server production environments.

University of Nairobi

Graduated Dec 2021

B.Sc. in Mathematics (Second Class Upper)

- Number Theory & Cryptography:** Applied finite fields and modular arithmetic to `secp256k1` curves and Taproot key path / script aggregation (BIP-340/341).
- Graph Theory & Algorithms:** Analyzed Dijkstra and min-cost flow algorithms for Lightning Network payment channel routing and balance state graphs.